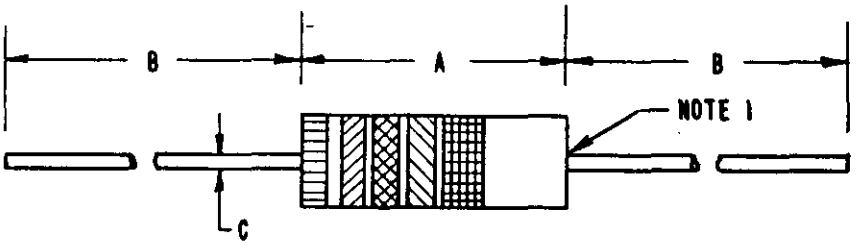
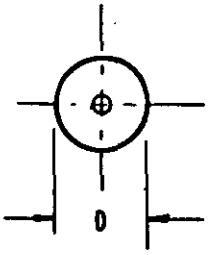
MIL-R-22684/6A(USAF)
7 September 1967
SUPERSEDING
22684/6 (USAF)
1 June 1965

MILITARY SPECIFICATION SHEET

RESISTORS, FIXED, FILM, INSULATED, STYLE RL20. . . . TX

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of MIL-R-22684.





	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.344	.416	8.74	10.57
B	1.375	1.625	34.92	41.28
C	.030	.034	.76	.86
D	.115	.161	2.92	4.09

NOTES:

1. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 150 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18mm) of the resistor body.
2. Metric equivalents are for general information only and are based upon 1 inch = 25.4mm.

FIGURE 1. DIMENSIONS

APPLICABLE DOCUMENT

MIL-STD-1276 Weldable Leads for Electronic Component Parts

REQUIREMENTS:

Dimensions and configuration: See figure 1.
Resistance values: See table I.

FSC 5905

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Table I. Part Number Designation

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL20S4R3GTX	4.3	G	S	1227	RL20S131GTX	130	G	S	1055
RL20S4R3JTX	4.3	J	S	1228	RL20S131JTX	130	J	S	1056
RL20S4R7GTX	4.7	G	S	1229	RL20S151GTX	150	G	S	1057
RL20S4R7JTX	4.7	J	S	1230	RL20S151JTX	150	J	S	1058
RL20S5R1GTX	5.1	G	S	1231	RL20S161GTX	160	G	S	1059
RL20S5R1JTX	5.1	J	S	1232	RL20S161JTX	160	J	S	1060
RL20S5R6GTX	5.6	G	S	1233	RL20S181GTX	180	G	S	1061
RL20S5R6JTX	5.6	J	S	1234	RL20S181JTX	180	J	S	1062
RL20S6R2GTX	6.2	G	S	1235	RL20S201GTX	200	G	S	1063
RL20S6R2JTX	6.2	J	S	1236	RL20S201JTX	200	J	S	1064
RL20S6R8GTX	6.8	G	S	1237	RL20S221GTX	220	G	S	1065
RL20S6R8JTX	6.8	J	S	1238	RL20S221JTX	220	J	S	1066
RL20S7R5GTX	7.5	G	S	1239	RL20S241GTX	240	G	S	1067
RL20S7R5JTX	7.5	J	S	1240	RL20S241JTX	240	J	S	1068
RL20S8R2GTX	8.2	G	S	1241	RL20S271GTX	270	G	S	1069
RL20S8R2JTX	8.2	J	S	1242	RL20S271JTX	270	J	S	1070
RL20S9R1GTX	9.1	G	S	1243	RL20S301GTX	300	G	S	1071
RL20S9R1JTX	9.1	J	S	1244	RL20S301JTX	300	J	S	1072
RL20S100GTX	10	G	S	1001	RL20S331GTX	330	G	S	1073
RL20S100JTX	10	J	S	1002	RL20S331JTX	330	J	S	1074
RL20S110GTX	11	G	S	1003	RL20S361GTX	360	G	S	1075
RL20S110JTX	11	J	S	1004	RL20S361JTX	360	J	S	1076
RL20S120GTX	12	G	S	1005	RL20S391GTX	390	G	S	1077
RL20S120JTX	12	J	S	1006	RL20S391JTX	390	J	S	1078
RL20S130GTX	13	G	S	1007	RL20S431GTX	430	G	S	1079
RL20S130JTX	13	J	S	1008	RL20S431JTX	430	J	S	1080
RL20S150GTX	15	G	S	1009	RL20S471GTX	470	G	S	1081
RL20S150JTX	15	J	S	1010	RL20S471JTX	470	J	S	1082
RL20S160GTX	16	G	S	1011	RL20S511GTX	510	G	S	1083
RL20S160JTX	16	J	S	1012	RL20S511JTX	510	J	S	1084
RL20S180GTX	18	G	S	1013	RL20S561GTX	560	G	S	1085
RL20S180JTX	18	J	S	1014	RL20S561JTX	560	J	S	1086
RL20S200GTX	20	G	S	1015	RL20S621GTX	620	G	S	1087
RL20S200JTX	20	J	S	1016	RL20S621JTX	620	J	S	1088
RL20S220GTX	22	G	S	1017	RL20S681GTX	680	G	S	1089
RL20S220JTX	22	J	S	1018	RL20S681JTX	680	J	S	1090
RL20S240GTX	24	G	S	1019	RL20S751GTX	750	G	S	1091
RL20S240JTX	24	J	S	1020	RL20S751JTX	750	J	S	1092
RL20S270GTX	27	G	S	1021	RL20S821GTX	820	G	S	1093
RL20S270JTX	27	J	S	1022	RL20S821JTX	820	J	S	1094
RL20S300GTX	30	G	S	1023	RL20S911GTX	910	G	S	1095
RL20S300JTX	30	J	S	1024	RL20S911JTX	910	J	S	1096
RL20S330GTX	33	G	S	1025	RL20S102GTX	1,000	G	S	1097
RL20S330JTX	33	J	S	1026	RL20S102JTX	1,000	J	S	1098
RL20S360GTX	36	G	S	1027	RL20S112GTX	1,100	G	S	1099
RL20S360JTX	36	J	S	1028	RL20S112JTX	1,100	J	S	1100
RL20S390GTX	39	G	S	1029	RL20S122GTX	1,200	G	S	1101
RL20S390JTX	39	J	S	1030	RL20S122JTX	1,200	J	S	1102
RL20S430GTX	43	G	S	1031	RL20S132GTX	1,300	G	S	1103
RL20S430JTX	43	J	S	1032	RL20S132JTX	1,300	J	S	1104
RL20S470GTX	47	G	S	1033	RL20S152GTX	1,500	G	S	1105
RL20S470JTX	47	J	S	1034	RL20S152JTX	1,500	J	S	1106
RL20S510GTX	51	G	S	1035	RL20S162GTX	1,600	G	S	1107
RL20S510JTX	51	J	S	1036	RL20S162JTX	1,600	J	S	1108
RL20S560GTX	56	G	S	1037	RL20S182GTX	1,800	G	S	1109
RL20S560JTX	56	J	S	1038	RL20S182JTX	1,800	J	S	1110
RL20S620GTX	62	G	S	1039	RL20S202GTX	2,000	G	S	1111
RL20S620JTX	62	J	S	1040	RL20S202JTX	2,000	J	S	1112
RL20S680GTX	68	G	S	1041	RL20S222GTX	2,200	G	S	1113
RL20S680JTX	68	J	S	1042	RL20S222JTX	2,200	J	S	1114
RL20S750GTX	75	G	S	1043	RL20S242GTX	2,400	G	S	1115
RL20S750JTX	75	J	S	1044	RL20S242JTX	2,400	J	S	1116
RL20S820GTX	82	G	S	1045	RL20S272GTX	2,700	G	S	1117
RL20S820JTX	82	J	S	1046	RL20S272JTX	2,700	J	S	1118
RL20S910GTX	91	G	S	1047	RL20S302GTX	3,000	G	S	1119
RL20S910JTX	91	J	S	1048	RL20S302JTX	3,000	J	S	1120
RL20S101GTX	100	G	S	1049	RL20S332GTX	3,300	G	S	1121
RL20S101JTX	100	J	S	1050	RL20S332JTX	3,300	J	S	1122
RL20S111GTX	110	G	S	1051	RL20S362GTX	3,600	G	S	1123
RL20S111JTX	110	J	S	1052	RL20S362JTX	3,600	J	S	1124
RL20S121GTX	120	G	S	1053	RL20S392GTX	3,900	G	S	1125
RL20S121JTX	120	J	S	1054	RL20S392JTX	3,900	J	S	1126

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Table I. Part Number Designation (Cont)

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL20S432GTX	4,300	G	S	1127	RL20S473GTX	47,000	G	S	1177
RL20S432JTX	4,300	J	S	1128	RL20S473JTX	47,000	J	S	1178
RL20S472GTX	4,700	G	S	1129	RL20S513GTX	51,000	G	S	1179
RL20S472JTX	4,700	J	S	1130	RL20S513JTX	51,000	J	S	1180
RL20S512GTX	5,100	G	S	1131	RL20S563GTX	56,000	G	S	1181
RL20S512JTX	5,100	J	S	1132	RL20S563JTX	56,000	J	S	1182
RL20S562GTX	5,600	G	S	1133	RL20S623GTX	62,000	G	S	1183
RL20S562JTX	5,600	J	S	1134	RL20S623JTX	62,000	J	S	1184
RL20S622GTX	6,200	G	S	1135	RL20S683GTX	68,000	G	S	1185
RL20S622JTX	6,200	J	S	1136	RL20S683JTX	68,000	J	S	1186
RL20S682GTX	6,800	G	S	1137	RL20S753GTX	75,000	G	S	1187
RL20S682JTX	6,800	J	S	1138	RL20S753JTX	75,000	J	S	1188
RL20S752GTX	7,500	G	S	1139	RL20S823GTX	82,000	G	S	1189
RL20S752JTX	7,500	J	S	1140	RL20S823JTX	82,000	J	S	1190
RL20S822GTX	8,200	G	S	1141	RL20S913GTX	91,000	G	S	1191
RL20S822JTX	8,200	J	S	1142	RL20S913JTX	91,000	J	S	1192
RL20S912GTX	9,100	G	S	1143		Megohm			
RL20S912JTX	9,100	J	S	1144	RL20S104GTX	0.10	G	S	1193
RL20S103GTX	10,000	G	S	1145	RL20S104JTX	0.10	J	S	1194
RL20S103JTX	10,000	J	S	1146	RL20S114GTX	0.11	G	S	1195
RL20S113GTX	11,000	G	S	1147	RL20S114JTX	0.11	J	S	1196
RL20S113JTX	11,000	J	S	1148	RL20S124GTX	0.12	G	S	1197
RL20S123GTX	12,000	G	S	1149	RL20S124JTX	0.12	J	S	1198
RL20S123JTX	12,000	J	S	1150	RL20S134GTX	0.13	G	S	1199
RL20S133GTX	13,000	G	S	1151	RL20S134JTX	0.13	J	S	1200
RL20S133JTX	13,000	J	S	1152	RL20S154GTX	0.15	G	S	1201
RL20S153GTX	15,000	G	S	1153	RL20S154JTX	0.15	J	S	1202
RL20S153JTX	15,000	J	S	1154	RL20S164GTX	0.16	G	S	1203
RL20S163GTX	16,000	G	S	1155	RL20S164JTX	0.16	J	S	1204
RL20S163JTX	16,000	J	S	1156	RL20S184GTX	0.18	G	S	1205
RL20S183GTX	18,000	G	S	1157	RL20S184JTX	0.18	J	S	1206
RL20S183JTX	18,000	J	S	1158	RL20S204GTX	0.20	G	S	1207
RL20S203GTX	20,000	G	S	1159	RL20S204JTX	0.20	J	S	1208
RL20S203JTX	20,000	J	S	1160	RL20S224GTX	0.22	G	S	1209
RL20S223GTX	22,000	G	S	1161	RL20S224JTX	0.22	J	S	1210
RL20S223JTX	22,000	J	S	1162	RL20S244GTX	0.24	G	S	1211
RL20S243GTX	24,000	G	S	1163	RL20S244JTX	0.24	J	S	1212
RL20S243JTX	24,000	J	S	1164	RL20S274GTX	0.27	G	S	1213
RL20S273GTX	27,000	G	S	1165	RL20S274JTX	0.27	J	S	1214
RL20S273JTX	27,000	J	S	1166	RL20S304GTX	0.30	G	S	1215
RL20S303GTX	30,000	G	S	1167	RL20S304JTX	0.30	J	S	1216
RL20S303JTX	30,000	J	S	1168	RL20S334GTX	0.33	G	S	1217
RL20S333GTX	33,000	G	S	1169	RL20S334JTX	0.33	J	S	1218
RL20S333JTX	33,000	J	S	1170	RL20S364GTX	0.36	G	S	1219
RL20S363GTX	36,000	G	S	1171	RL20S364JTX	0.36	J	S	1220
RL20S363JTX	36,000	J	S	1172	RL20S394GTX	0.39	G	S	1221
RL20S393GTX	39,000	G	S	1173	RL20S394JTX	0.39	J	S	1222
RL20S393JTX	39,000	J	S	1174	RL20S434GTX	0.43	G	S	1223
RL20S433GTX	43,000	G	S	1175	RL20S434JTX	0.43	J	S	1224
RL20S433JTX	43,000	J	S	1176	RL20S474GTX	0.47	G	S	1225
					RL20S474JTX	0.47	J	S	1226

Resistance tolerances: See table I.
Terminal type: See table I.
Minimum resistance value: 4.3 ohms.
Maximum resistance value: 0.47 megohm.

Maximum continuous working voltage: 350 volts.

Power rating: 1/2 watt at 70°C.

Temperature cycling: Method 102 of MIL-STD-202, condition D.

Change in resistance shall not exceed ±(1 percent + 0.05 ohm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Low temperature operation:
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Short time overload:
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Terminal strength: Method 211 of MIL-STD-202.
Condition A-Pull test 5 pounds.
Condition D—Twist test: rotated 360° for 3 rotations. Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Dielectric withstanding voltage:
Methods 301 and 105 of MIL-STD-202.
Atmospheric pressure—700 volts rms.
Barometric pressure—400 volts rms.

Insulation resistance: Method 302 of MIL-STD-202. Condition A or B, whichever is practicable—Insulation resistance shall be not less than 100 megohms.

Resistance to soldering heat: Method 210 of MIL-STD-202.
Condition A—Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Moisture resistance: Method 106 of MIL-STD-202. Change in resistance shall not exceed 1.5 percent. Insulation resistance shall be 100 megohms, minimum.

Resistance temperature characteristic: Method 304 of MIL-STD-202. Change in resistance shall not exceed ±0.020 percent per degree C.

Life: Method 108 of MIL-STD-202.
Condition D—Change in resistance shall not exceed 2.0 percent.

Shock, medium impact: Method 205 of MIL-STD-202.
Condition C—Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Vibration, high frequency: Method 204 of MIL-STD-202.
Condition D—Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Solderability: Method 208 of MIL-STD-202.

Lead material: Type C, in accordance with MIL-STD-1276.

Lead coating: Tin-lead (Sn 30 to 70 percent) 0.0005- to 0.0015-inch thickness. The maximum thickness shall not exceed twice the minimum thickness at any cross section. The leads shall be free of nonconductive and foreign material beyond the maximum body dimensions.

QUALITY ASSURANCE PROVISIONS

Sampling and inspection: MIL-R-22684 and as specified herein.

Qualification inspection: MIL-R-22684 on the respective not-TX version. (The TX version shall be procured from sources listed in QPL-22684 for the respective non-TX version.)

Quality conformance inspection: Inspection of product for delivery: Groups A, B,

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

and C inspections. Delivery of shipment shall not be delayed pending completion of group C testing.

Group B inspection: MIL-R-22684, except that the AQL shall be 2.5 (percent defective).

Group C inspection: MIL-R-22684. Successful performance of group C inspection for the respective non-TX version will suffice to demonstrate conformance to this requirement for TX version.

Power conditioning

Measurement before conditioning: D-c resistance measured in accordance with paragraph entitled "DC Resistance" of MIL-R-22684.

Conditioning procedures: Each resistor supplied under this specification shall be subjected to power conditioning for a period of 24 hours at 150 percent rated power. Mounting shall be such that 25° ±2°C ambient air can circulate around the resistors by means of normal convection. The maximum applied voltage shall not exceed 350V rms.

Measurement after conditioning: After power conditioning, the resistors shall be stabilized at room conditions for a minimum of 2 hours before the d-c resistance measurements are made. Resistors that are outside of the ±2 percent tolerance limit and those that have changed more than 0.5 percent due to the conditioning shall be removed from the lot. When the combined quantity removed exceeds 5 percent of the lot, the entire lot shall be rejected. The supplier shall have a complete record available of all test data obtained during power conditioning testing (see ordering data).

TX identification: Resistors procured to, and meeting all of the criteria specified herein shall bear a green colored band in place of the band "E" in accordance with MIL-STD-221. The symbol "TX" shall be printed at one of the locations specified below:

a. Preferred: On the green colored band described above.

b. Alternate: On the packing which packaged the items in groups.

In the event the resistor sample fails to meet the requirements of MIL-R-22684 and this specification sheet, the manufacturer shall remove the green band from the sample tested and also from all the resistors represented by the sample.

Qualification: The activity responsible for the Qualified Products List is the Systems Engineering Group, Attn: SEPSE, Wright-Patterson Air Force Base, Ohio 45433. However, information pertaining to the qualification of products may be obtained from the Defense Electronics Supply Center, Attn: DESC-EP, 1507 Wilmington Pike, Dayton, Ohio 45420.

Ordering data: The procurement document should specify whether the power conditioning data is to be supplied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Part number: Consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

M22684/06-1001

Basic	Dash number
Document	from
Number	Table I.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 17, 85

Proj. No. 5905-F225

*U.S. GOVERNMENT PRINTING OFFICE: 1967-301-512/1676

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.